Aoran Wang

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Education

University of Luxembourg — Esch-sur-Alzette, Luxembourg
PhD in Computer Science (Focus: Machine Learning for Dynamical Systems) — *Dec. 2020 – Nov. 2024*

Karlsruhe Institute of Technology — Karlsruhe, Germany
Master of Science (Major: Autonomous Driving); GPA: 3.9/4.0 — *Oct. 2015 – Sep. 2019*

Tongji University — Shanghai, China
Bachelor of Engineering (Major: Structural Mechanics); GPA: 3.9/4.0 — *Sep. 2011 – Jun. 2015*

Research Interests

- AI for Science and multimodal foundation models.
- Scientific and formal reasoning, with an emphasis on equation discovery and visual-to-symbolic inference.
- Structural inference and graph learning for dynamical systems.

Skills

Languages: Python (advanced), C/C++ (advanced), HTML/CSS, MATLAB, Visual Basic, LaTeX.
Developer Tools: Linux, Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, Slurm.
Libraries: PyTorch, TensorFlow, JAX, Scikit-Learn, Jupyter, pandas, NumPy, Matplotlib, CUDA.
Techniques: Multimodal Models, Scientific Reasoning, Equation Discovery, Structural Inference, Graph Neural Networks, Variational Autoencoders, Active Learning, Generative Flow Networks, Dynamical Systems.

Publications

- Liu, J., Lai, S., Li, P., Yu, D., Zhou, W., Zhou, Y., Xia, P., Wang, Z., Chen, X., Tang, S., Bai, L., Ouyang, W., Ding, M., Yao, H., and Wang, A. (2026). Mimicking the Physicist's Eye: A VLM-centric Approach for Physics Formula Discovery. NeurIPS 2026. Accepted. (Link to paper.)
- Su, E., Wu, J., Zhang, J., Yu, Q., Tang, C., Li, P., Wang, L., Wang, A., Ma, X., Tang, S., Wang, Y., and Li, H. (2026). S3Mem: Structured Spatiotemporal Scene-Event Memory for Long-Horizon Interactive Question Answering. EMNLP 2026, Main Conference. Accepted. (Link to paper.)
- Wang, A. and Pang, J. (2026). Harnessing Generative Flow Networks for Effective Structural Inference. Machine Learning, 115(5), Article 113. (Link to paper.)
- Wan, H., Yang, C., Yu, J., Tu, M., Lu, J., Yu, D., Cao, J., Gao, B., Xie, J., Wang, A., Zhang, W., Torr, P., and Zhou, D. (2026). Deep Research Arena: The First Exam of LLMs' Research Abilities via Seminar-Grounded Tasks. AAAI 2026, 40(39), pp. 33341–33349. (Link to paper.)
- Tong, T.P., Wang, A., Panagopoulos, G., and Pang, J. (2026). Kolmogorov-Arnold Network for Gene Regulatory Network Inference. CMSB 2025 (LNCS), 15959, pp. 221–239. (Link to paper.)
- Tong, T.P., Wang, A., Panagopoulos, G., and Pang, J. (2026). Integrating Optimal Transport and Structural Inference Models for GRN Inference from Single-cell Data. ECML PKDD 2024 Workshops (CCIS), 2560, pp. 257–271. (Link to paper.)
- Dai, X., Yang, K., Revolinsky, J., Guo, K., Wang, A., Zhang, B., and Tang, J. (2025). From Sequence to Structure: Uncovering Substructure Reasoning in Transformers. NeurIPS 2025, 38. (Link to paper.)
- Wang, A., Dai, X., and Pang, J. (2025). Guided Structural Inference: Leveraging Priors with Soft Gating Mechanisms. ICML 2025, 267, pp. 62761–62783. (Link to paper.)
- Wang, A. and Pang, J. (2024). Structural Inference of Dynamical Systems with Conjoined State Space Models. NeurIPS 2024. (Link to paper.)
- Wang, A., Tong, T.P., Mizera, A., and Pang, J. (2024). Benchmarking Structural Inference Methods for Interacting Dynamical Systems with Synthetic Data. NeurIPS 2024 Datasets and Benchmarks Track. (Link to paper.)

- Wang, A. and Pang, J. (2024). Structural Inference with Dynamics Encoding and Partial Correlation Coefficients. ICLR 2024. (Link to paper.)
- Wang, A., Tong, T.P., and Pang, J. (2023). Effective and Efficient Structural Inference with Reservoir Computing. ICML 2023, 202, pp. 36391–36410. (Link to paper.)
- Wang, A. and Pang, J. (2023). Active Learning based Structural Inference. ICML 2023, 202, pp. 36224–36245. (Link to paper.)
- Wang, A. and Pang, J. (2022). Iterative Structural Inference of Directed Graphs. NeurIPS 2022, 35, pp. 8717–8730. (Link to paper.)
- Hu, H., Wang, A., Sons, M., and Lauer, M. (2020). ViPNet: An End-to-End 6D Visual Camera Pose Regression Network. ITSC 2020, pp. 1-7. (Link to paper.)

Preprints

- Li, P., Zhang, J., Su, E., Long, Y., Zeng, Z., Liu, X., Zhang, H., Yu, J., Tang, S., Wang, A., and Chen, X. (2026). Coupled Harness–Model Evolution for Scientific Equation Discovery. arXiv:2610.17740. (Link to preprint.)
- Su, E., Liu, X., Wang, Y., Zhong, B., Li, P., Wu, J., Li, M., Ma, X., Wang, A., Hong, L., Ouyang, W., Tang, C., Li, H., and Tang, S. (2026). PROTEAN: Beyond De Novo Generation - Can LLM Agents Perform Text-Guided Protein Structure Editing? arXiv:2610.17765. (Link to preprint.)
- Su, E., Li, P., Hua, Z., Yu, Q., Wu, J., Tang, C., Wang, Y., Wang, L., Xiang, J., Ma, X., Chen, X., Tang, S., Li, H., Chen, X., and Wang, A. (2026). BRIDGE: Selective Boundary Repair with Dependency Graph Evidence in Multi-Agent Systems. arXiv:2610.17805. (Link to preprint.)
- Tang, C., Wang, Y., Wu, J., . . . , Wang, A., . . . , Bai, L. (2026). Accurate, Interdisciplinary and Transparent Structure-property Understanding with Deep Native Structural Reasoning. arXiv:2607.07708. (Link to preprint.)
- Li, P., Zhang, J., Long, Y., Liu, X., Zhou, W., Su, E., Zeng, Z., Liu, J., Liu, J., Yu, J., Liu, L., Torr, P., Tang, S., Wang, A., and Chen, X. (2026). Hidden in Plain Sight: Visual-to-Symbolic Analytical Solution Inference from Field Visualizations. arXiv:2604.08863. (Link to preprint.)
- Su, E., Wu, J., Tang, C., Wang, L., Li, P., Wang, A., Zhang, J., Wang, Y., Meng, Y., Ma, X., Tang, S., and Li, H. (2026). SciIF: Benchmarking Scientific Instruction Following Towards Rigorous Scientific Intelligence. arXiv:2601.04770. (Link to preprint.)
- Hu, M., Ma, C., Li, W., . . . , Wang, A., . . . , Zhou, B. (2025). A Survey of Scientific Large Language Models: From Data Foundations to Agent Frontiers. arXiv:2508.21148. (Link to preprint.)
- Bai, L., Cai, Z., Cao, Y., . . . , Wang, A., . . . , Zou, Y. (2025). Intern-S1: A Scientific Multimodal Foundation Model. arXiv:2508.15763. (Link to preprint.)
- Wang, L., Su, E., Liu, J., Li, P., Xiao, J., Zhang, W., Dai, X., Chen, X., Meng, Y., Bai, L., Ouyang, W., Tang, S., Wang, A., and Ma, X. (2025). PhysUniBench: A Multi-Modal Physics Reasoning Benchmark at Undergraduate Level. arXiv:2506.17667. (Link to preprint.)
- Liang, H., Wang, A., Zhou, J., Jin, X., Gao, C., and Wang, J. (2025). Examining the Source of Defects from a Mechanical Perspective for 3D Anomaly Detection. arXiv:2505.05901. (Link to preprint.)

Projects

Multimodal Models and Scientific Reasoning *2025 – Present*

- Research on scientific multimodal foundation models and native structural reasoning across scientific domains.
- Visual-to-symbolic discovery of physical laws with VIPER-R1.
- Physics reasoning evaluation with PhysUniBench and scientific instruction following with SciIF.

LLM4EDUKG | *Python, Git, Unix Shell, VS Code* *Mar. 2024 – Jun. 2024*

- Curated four educational knowledge graphs with knowledge components and prerequisite relationships.
- Evaluated pre-trained language models on prerequisite relationships and multi-hop reasoning.
- Released the dataset, evaluation pipeline, and results via GitHub and the project website.

StructInfer | *Python, Slurm, Git, Unix Shell, VS Code* *Dec. 2022 – Apr. 2024*

- Generated observational trajectories for dynamical systems across multiple scientific domains.
- Evaluated 13 structural inference methods for accuracy, scalability, data efficiency, and robustness.
- Released datasets, code, and benchmark results via GitHub and the project website.

Experience

Shanghai AI Laboratory Shanghai, China

Researcher *May 2025 – Present*

- Conduct research in AI for Science, focusing on multimodal models and scientific reasoning.
- Investigate equation discovery, visual-to-symbolic reasoning, and structural inference of dynamical systems.
- Contribute to scientific foundation models, native structural reasoning, and physics reasoning benchmarks.

Squirrel AI Hybrid: Shanghai and remote

Research Intern *Feb. 2024 – Apr. 2025*

- Developed recommendation models for K-12 education using spatiotemporal graphs and integrated databases.
- Addressed missing educational records using spatiotemporal similarities.
- Led LLM4EDUKG to evaluate large language model reasoning on educational knowledge graphs.
- Investigated GraphRAG for incorporating domain knowledge into LLMs and simulated student behavior.

University of Luxembourg Esch-sur-Alzette, Luxembourg

Doctoral Researcher (PhD Student) *Dec. 2020 – Nov. 2024*

- Researched deep learning methods for structural inference in dynamical systems.
- Developed StructInfer, an open benchmark for reproducible evaluation across scientific domains.
- Published at NeurIPS, ICML, and ICLR, and reviewed submissions for leading AI conferences.

Karlsruhe Institute of Technology Karlsruhe, Germany

Student Assistant *Dec. 2019 – Mar. 2020*

- Developed monocular visual localization methods using graph neural networks and OpenCV.
- Published visual localization research in a peer-reviewed conference.
- Investigated hierarchical and combinatorial approaches to visual localization.

Robert Bosch GmbH Renningen, Germany

Research Intern *Apr. 2018 – Aug. 2018*

- Delivered an AUTOSAR tutorial and communicated the work to the research team and director.
- Developed an extended Kalman filter diagnosis system for the electric propulsion of an autonomous vehicle prototype.
- Explored AI methods for diagnostics in autonomous driving.

Honors and Awards

- College-level Excellence, 2011-2015
- Outstanding Graduates of Tongji University, 2015
- Travel Grant, NeurIPS 2022, ICML 2023, ICLR 2024
- Top Reviewer, NeurIPS 2023, NeurIPS 2024
- DAAD AInet Fellow

Interests

Indoor Sports: Fitness, Badminton, Table Tennis.

Outdoor Sports: Snorkeling, Sky Diving, Hiking, Skiing, Surfing.

Arts: Photography, Classical Music, Jazz Music.